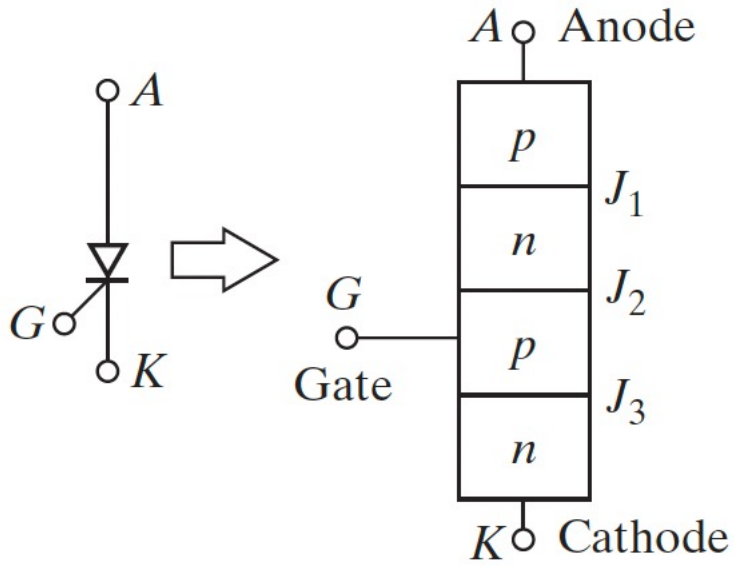
Lecture 7

Silicon Controlled Rectifier : Thyristor

- Thyristors are a family of power semiconductor devices. Thyristors are used extensively in power electronic circuits. They are operated as bistable switches, operating from nonconducting state to conducting state.
- Thyristors can be assumed as ideal switches for many applications, but the practical thyristors exhibit certain characteristics and limitations.
- Conventional thyristors are designed without gate-controlled turn-off capability, in which case the thyristor can recover from its conducting state to a nonconducting state only when the current is brought to zero by other means.
- Gate turn-off thyristors (GTOs) are designed to have both controlled turn-on and turn-off capability.
- Compared to transistors, thyristors have lower on-state conduction losses and higher power handling capability. On the other hand, transistors generally have superior switching performances in terms of faster switching speed and lower switching losses

- A thyristor is a four-layer semiconductor device of PNPN structure with three pn junctions. It has three terminals: anode, cathode, and gate.



Animation Video: https://www.youtube.com/watch?v=0AgPUikpvpM

SCR IV Characteristics

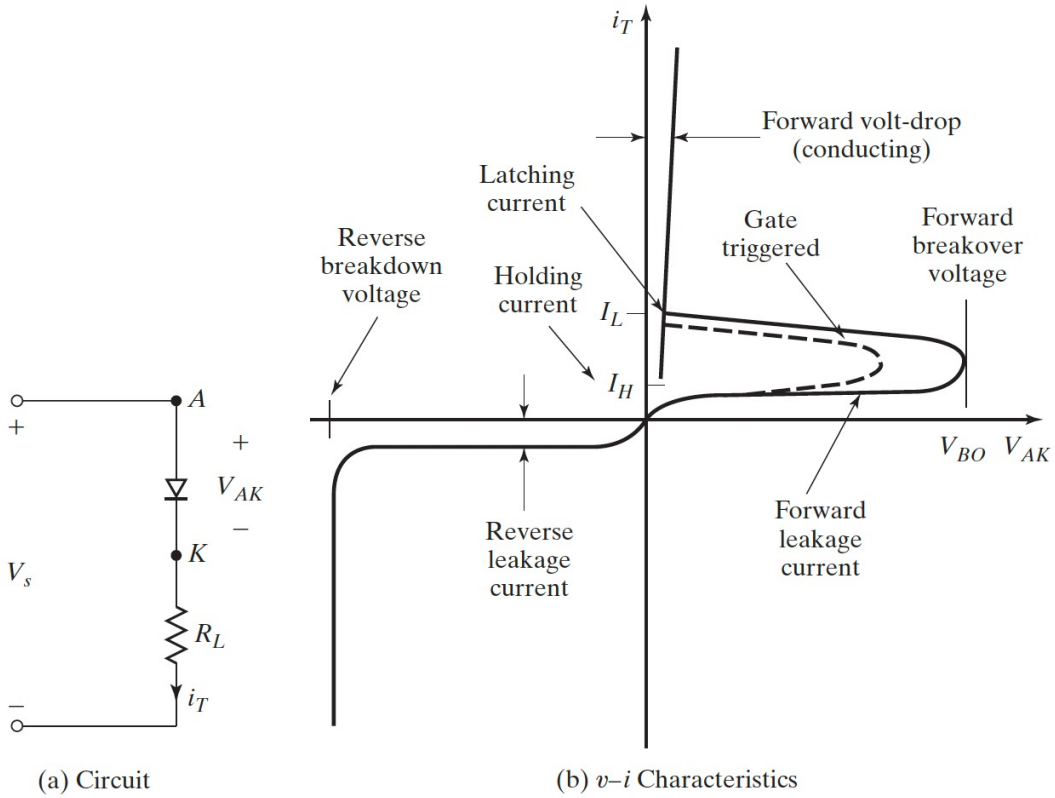
- When the anode voltage is made positive with respect to the cathode, the junctions J1 and J3 are forward biased. The junction J2 is reverse biased, and only a small leakage current flows from anode to cathode. The thyristor is then said to be in the forward blocking, or off-state, condition and the leakage current is known as off-state current.
- If the anode-to-cathode voltage V_{AK} is increased to a sufficiently large value, the reverse-biased junction J2 breaks. It is known as avalanche breakdown and the corresponding voltage is called forward breakdown voltage V_{BO}.
- Because the other junctions J1 and J3 are already forward biased, there is free movement of carriers across all three junctions, resulting in a large forward anode current. The device is then in a conducting state, or on-state. The voltage drop would be due to the ohmic drop in the four layers and it is small, typically, 1 V.

SCR IV Characteristics

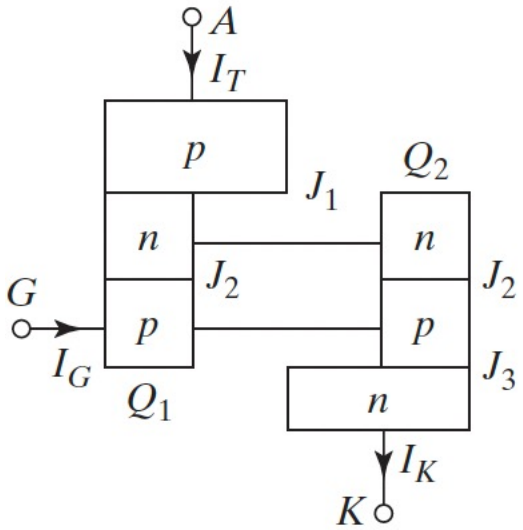
- **Latching current:** The anode current must be more than a value known as latching current I_L to maintain the required amount of carrier flow across the junction; otherwise, the device reverts to the blocking condition as the V_{AK} is reduced. Latching current I_L is the minimum anode current required to maintain the thyristor in the on-state immediately after a thyristor has been turned on and the gate signal has been removed.
- Once a thyristor conducts, it behaves like a conducting diode and there is no control over the device. The device continues to conduct because there is no depletion layer on the junction J2 due to free movements of carriers.
- **Holding current**: if the forward anode current is reduced below a level known as the holding current I_H, a depletion region develops around junction J2 due to the reduced number of carriers and the thyristor is in the blocking state. The holding current is on the order of milliamperes and is less than the latching current I_L. That is, $I_L > I_H$. Holding current I_H is the minimum anode current to maintain the thyristor in the on-state. The holding current is less than the latching current.

SCR IV Characteristics

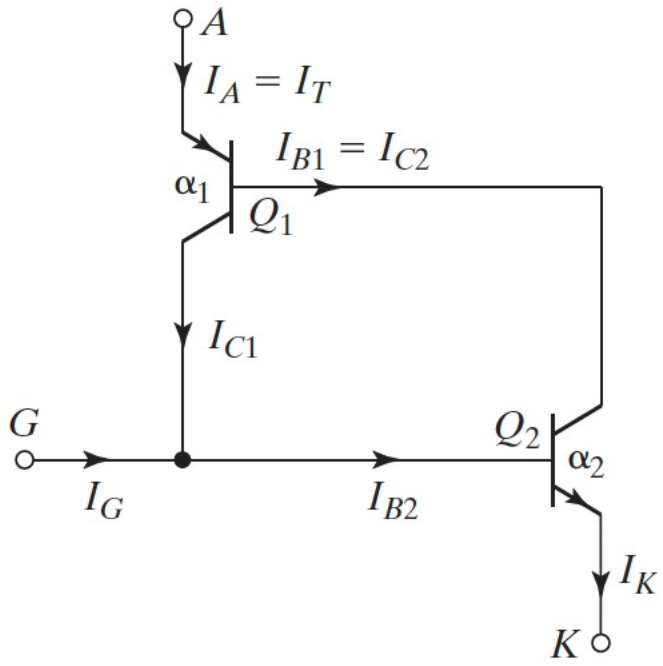
- When the cathode voltage is positive with respect to the anode, the junction J2 is forward biased but junctions J1 and J3 are reverse biased. This is like two series-connected diodes with reverse voltage across them. The thyristor is in the reverse blocking state and a reverse leakage current, known as reverse current I_R, flows through the device.
- A thyristor can be turned on by increasing the forward voltage V_{AK} beyond V_{BO}, but such a turn-on could be destructive. In practice, the forward voltage is maintained below V_{BO} and the thyristor is turned on by applying a positive voltage between its gate and cathode.



Two Transistor Model of Thyristor



(a) Basic structure



(b) Equivalent circuit

The regenerative or latching action due to a positive feedback can be demonstrated by using a two-transistor model of thyristor. A thyristor can be considered as two complementary transistors, one PNP-transistor, Q1, and other NPN-transistor, Q2.

The collector current I_C of a thyristor is related, in general, to the emitter current I_E and the leakage current of the collector–base junction, I_{CBO}, as

$$I_C = \alpha I_E + I_{CBO}$$

$$I_{C1} = \alpha_1 I_A + I_{CBO1}$$

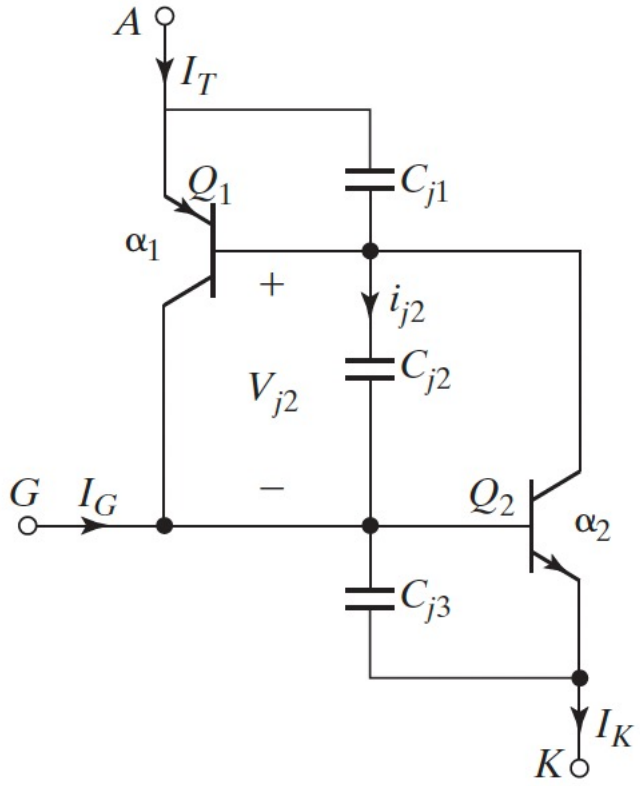
$$I_{C2} = \alpha_2 I_K + I_{CBO2}$$

$$I_A = I_{C1} + I_{C2} = \alpha_1 I_A + I_{CBO1} + \alpha_2 I_K + I_{CBO2}$$

$$I_K = I_A + I_G$$

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

Thyristor Transient Model



The capacitances of the pn-junctions, as shown in Figure, influence the characteristics of the thyristor. If a thyristor is in a blocking state, a rapidly rising voltage applied across the device would cause high current flow through the junction capacitors. The current through capacitor C_{j2} can be expressed as

$$i_{j2} = \frac{d(q_{j2})}{dt} = \frac{d}{dt}(C_{j2}V_{j2}) = V_{j2}\frac{dC_{j2}}{dt} + C_{j2}\frac{dV_{j2}}{dt} \qquad (9.6)$$

where C_{j2} and V_{j2} are the capacitance and voltage of junction J_2, respectively, Transistor Q_{j2} is the charge in the junction. If the rate of rise of voltage dv/dt is large, then i_{j2} would be large and this would result in increased leakage currents I_{CBO1} and I_{CBO2}.

Thyristor Turn on

$$i_{j2} = \frac{d(q_{j2})}{dt} = \frac{d}{dt}(C_{j2}V_{j2}) = V_{j2}\frac{dC_{j2}}{dt} + C_{j2}\frac{dV_{j2}}{dt} \quad (9.6)$$

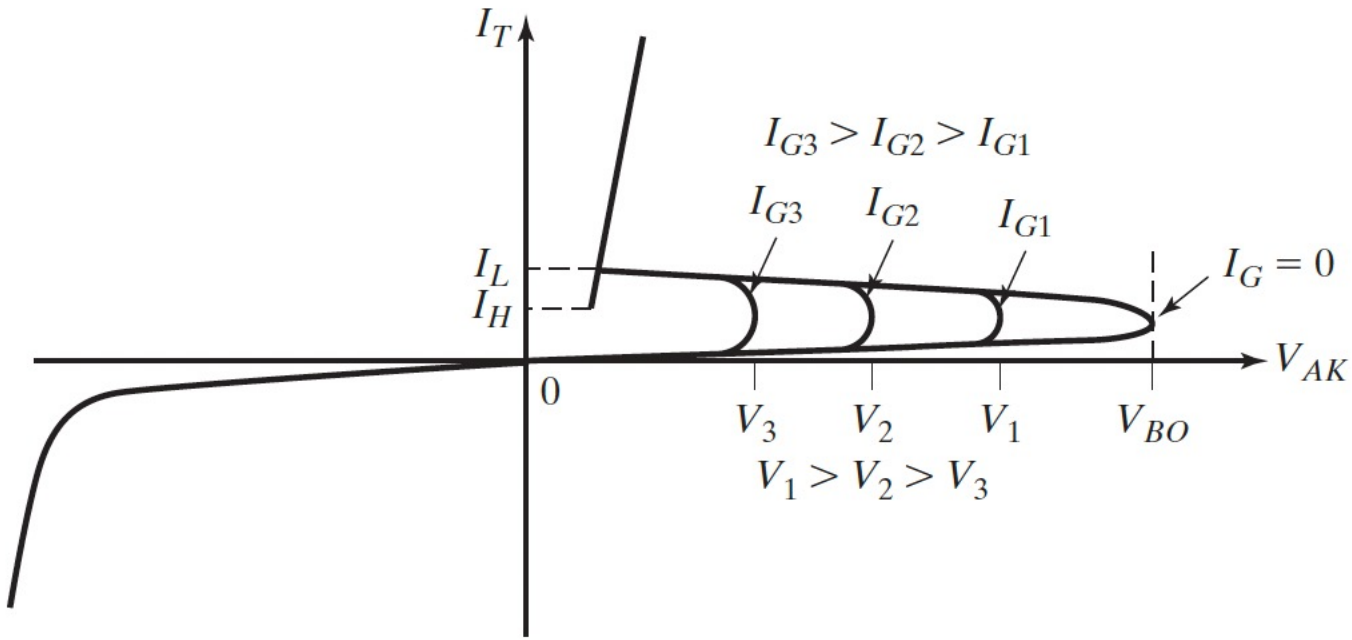
Thermals. If the temperature of a thyristor is high, there is an increase in the number of electron–hole pairs, which increases the leakage currents. This increase in currents causes α_1 and α_2 to increase. Due to the regenerative action, $(\alpha_1 + \alpha_2)$ may tend to unity and the thyristor may be turned on. This type of turn-on may cause thermal runaway and is normally avoided.

Light. If light is allowed to strike the junctions of a thyristor, the electron–hole pairs increase; and the thyristor may be turned on. The light-activated thyristors are turned on by allowing light to strike the silicon wafers.

High voltage. If the forward anode-to-cathode voltage is greater than the forward breakdown voltage V_{BO}, sufficient leakage current flows to initiate regenerative turn-on. This type of turn-on may be destructive and should be avoided.

dv/dt. It can be noted from Eq. (9.6) that if the rate of rise of the anode–cathode voltage is high, the charging current of the capacitive junctions may be sufficient enough to turn on the thyristor. A high value of charging current may damage the thyristor; and the device must be protected against high *dv*/*dt*. The manufacturers specify the maximum allowable *dv*/*dt* of thyristors.

Gate current. If a thyristor is forward biased, the injection of gate current by applying positive gate voltage between the gate and cathode terminals turns on the thyristor. As the gate current is increased, the forward blocking voltage is decreased,

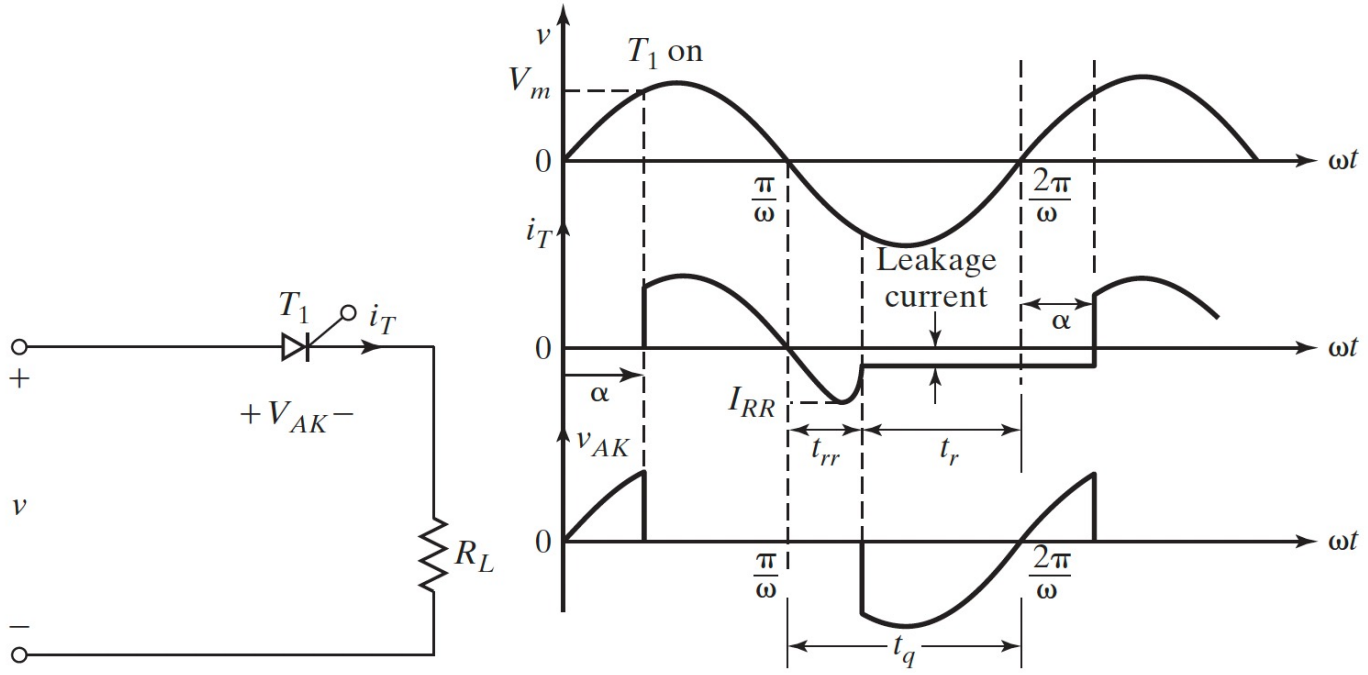


Effects of gate current on forward blocking voltage.

Turning Off Thyristor

- A thyristor that is in the on-state can be turned off by reducing the forward current to a level below the holding current I_H. There are various techniques for turning off a thyristor. In all the commutation techniques, the anode current is maintained below the holding current for a sufficiently long time, so that all the excess carriers in the four layers are swept out or recombined.
- Due to two outer pn-junctions, J1 and J3, the turn-off characteristics would be similar to that of a diode, exhibiting reverse recovery time t_{rr} and peak reverse recovery current I_{RR}. I_{RR} can be much greater than the normal reverse blocking current I_R. In a line-commutated converter circuit where the input voltage is alternating, a reverse voltage appears across the thyristor immediately after the forward current goes through the zero value. This reverse voltage accelerates the turn-off process, by sweeping out the excess carriers from pn-junctions J_1 and J_3.

- The turn-off time tq is the sum of reverse recovery time t_{rr} and recombination time t_{rc}. At the end of turn-off, a depletion layer develops across junction J2 and the thyristor recovers its ability to withstand forward voltage. In all the commutation techniques, a reverse voltage is applied across the thyristor during the turn-off process.



(a) Line-commutated thyristor circuit

Thank you for your attention

Next Class

Forced Commutated Thyristor Circuit

